DHEERAJ AMARAVADI

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PROFESSIONAL SUMMARY

Applied AI Engineer with hands-on experience building production-oriented AI systems, enterprise AI systems, RAG-based knowledge retrieval, LLM-powered workflows, MCP servers and backend services. Experienced in translating engineering requirements into reusable Agent Skills, integrating enterprise knowledge and capabilities with AI agents and building scalable backend services for agent orchestration and tool execution. Strong software engineering background in Python, C#, JavaScript, REST APIs, with additional experience in machine learning, deep learning, prompt engineering and LLM evaluation. Experienced in applying specification-driven development and collaboration with cross-functional teams to deliver AI solutions for real world engineering workflows.

WORK EXPERIENCE

Software Engineer | Tektronix | April 2025 – Present

AI Engineering

- Developing production-oriented Generative AI application and AI Agent Skills for an internal enterprise AI assistant platform.
- Implemented RAG-based knowledge retrieval by structuring application and backend knowledge using Open Knowledge Format (OKF) and integrating contextual knowledge into reusable Agent Skills.
- Leveraged metadata, contextual information and structured knowledge to enable Agent Skills to identify and retrieve relevant enterprise knowledge during task execution.
- Building reusable Model Context Protocol (MCP) servers to expose enterprise capabilities and tools to AI agents.
- Designing backend architecture for AI agent workflows, supporting agent orchestration, knowledge retrieval, tool execution and enterprise integrations.
- Developing LLM-powered workflow automation to support engineering processes and reduce manual intervention.
- Applied Specification-Driven Development using GitHub Spec Kit to translate requirements into structured specifications, implementation plans and production ready AI applications.
- Built scalable backend services supporting agent execution, enterprise tool integration and AI assisted engineering workflows.
- Collaborated with cross functional engineering teams to understand requirements, integrate AI capabilities and deliver production-oriented solutions.

Oscilloscope Automation & Reporting Framework

- Designed and developed an end-to-end C# automation framework for configuring oscilloscope parameters (record length, sample rate, clock recovery, REF Waveforms, measurement settings), executing measurements, validating outputs and generating reports through SCPI based instrument control commands.
- Built a golden result comparison engine to detect measurement deviations across legacy and alternate instrument versions, enabling automated validation of instrument outputs.
- Developed a WPF-based workflow that unified configuration, measurement execution, validation and reporting, eliminating manual execution steps.

Junior Developer | ILaqa Technologies (Remote) | July 2024 – February 2025

- Sole backend developer for an on-demand home services platform, responsible for building and delivering all backend services to production.
- Built 15+ RESTful APIs using Node.js and Express, implementing business logic including service listings with area-based filtering and pagination, referral management and service request status flow management.
- Designed MongoDB schemas for users, services, areas and referrals with cross collection lookups to support relational data fetching across the platform.
- Implemented secure authentication using AWS Cognito with role-based access control, JWT-based endpoint protection and token refresh logic, managed profile AWS S3 image uploads and built production-grade backend infrastructure with centralized error handling, request validation and logging.

Freelance LLM Model Evaluator & Code Reviewer | Outlier.ai (Remote) | May 2024 – June 2024

- Evaluated and ranked LLM-generated code for correctness and quality.
- Designed prompts to elicit accurate and task-specific model responses.
- Provided structured feedback on generated code, gaining hands-on experience in prompt engineering and model evaluation.

ACADEMIC PROJECTS

Crop Recommendation System

- Built a 97% accurate ML model predicting suitable crops based on soil and environmental parameters.
- Utilized Random Forest, Decision Tree, Logistic Regression, and Gaussian Naive Bayes algorithms.
- Tech Stack: Python, Scikit-learn, Flask
  GitHub: https://github.com/AMARAVADIDHEERAJ/Crop-Recommendation-System

Network Intrusion Detection System

- Developed a deep learning-based IDS using 1D CNN and Adam optimizer, balanced imbalanced datasets using SMOTE, achieving 98.59% accuracy.
- Tech Stack: Python, TensorFlow, Pandas, NumPy
- GitHub: https://github.com/AMARAVADIDHEERAJ/Network-Intrusion-Detection-System-Using-Deep-Learning

TECHNICAL SKILLS

AI & Generative AI: Generative AI, Large Language Models (LLMs), AI Agents, Agent Skills, Retrieval Augmented Generation (RAG), Open Knowledge Format (OKF), Model Context Protocol (MCP), LLM Integration, Prompt Engineering, LLM Evaluation
Programming Languages: Python, C#, JavaScript, SQL
AI/ML: Machine Learning, Deep Learning, TensorFlow, Scikit-learn, Neural Networks
Backend & APIs: REST APIs, Node.js, Express.js, API Integration, MongoDB, Microservices, MySQL
Cloud & Engineering: AWS Cognito, AWS S3, Postman, WPF, Object-Oriented Programming, Specification-Driven Development, GitHub Spec Kit

CERTIFICATIONS

- Oracle Cloud Infrastructure 2024 Generative AI Professional | Oracle | certification
- Intermediate Machine Learning (Kaggle Certification) | certification

EDUCATION DETAILS

B. Tech | CSE with specialization in Artificial Intelligence and Machine Learning | 2020 - 2024
Vellore Institute of Technology, Chennai | **CGPA: 7.76**

LEADERSHIP

- **Secretary - National Service Scheme (NSS) VIT CHENNAI -** Organized and led community wellness initiatives, including health camps and awareness drives, impacting 1000+ individuals and strengthening community engagement.